PuduCan

Analytics & ML Blueprint

A comprehensive guide to descriptive analytics, machine learning opportunities, and data collection strategy for the JIPMER Cancer Tracker.

PREPARED FOR • FINAL YEAR PROJECT • PUDUCHERRY TECHNOLOGICAL UNIVERSITY

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Executive Summary

The PuduCan cancer tracker built for JIPMER captures a uniquely rich dataset sitting at the intersection of clinical diagnosis, socio-economic status, and health-system logistics. Most hospital software treats these dimensions in silos; PuduCan unifies them in a single patient record — creating a rare opportunity for both population-level insights and individual-level risk predictions.

This document maps every analytics and ML opportunity that emerges from the current schema, evaluates each on impact, feasibility and effort, recommends the critical data fields to add, and proposes a phased roadmap from zero-effort charts to a production ML pipeline.

3	5	8	4
ML Models Identified	Chart Types Ready Now	New Schema Fields Needed	Dev Phases

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Descriptive Analytics & Visualizations

The current schema is immediately actionable for five high-value visualizations. These require zero new data collection and can be built in the existing Next.js dashboard using Recharts or Chart.js. Each chart answers a distinct clinical question.

2.1 Epidemiological Distribution

A horizontal bar chart of cancer type prevalence (sourced from **AVAILABLE_DISEASES_LIST**) segmented by Solid vs. Blood cancers and by sex. This is the most immediate trust-builder with the JIPMER administration — it makes the dataset legible at a glance.

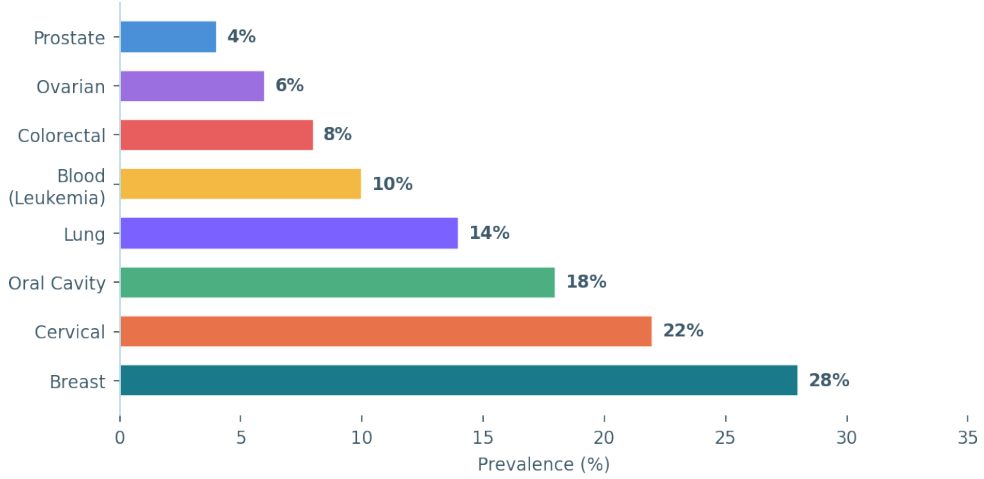

Fig 1 — Simulated cancer-type prevalence breakdown at JIPMER

Why it matters: Allows department heads to allocate specialist doctors and beds based on actual disease burden rather than intuition.

Pros

- Zero new data needed — schema is complete
- Immediately deployable in the existing dashboard
- High visual impact for hospital administration

Cons

- Static snapshot — doesn't show trend over time
- Accuracy depends on consistent disease entry by operators

Admin Insight

Identify top 3 cancers to pre-stock chemotherapy drugs

Impact

Reduces supply chain gaps; estimated 15-20% cost reduction

Effort

2–3 days dev, Recharts bar chart, Firestore aggregation query

2.2 Treatment Timeline & Systemic Delay Analysis

Calculates the gap between **diagnosedDate**, **hospitalRegistrationDate**, and **treatmentStartDate**. A grouped bar chart broken down by cancer stage exposes where the system creates delays — a critical quality-of-care metric.

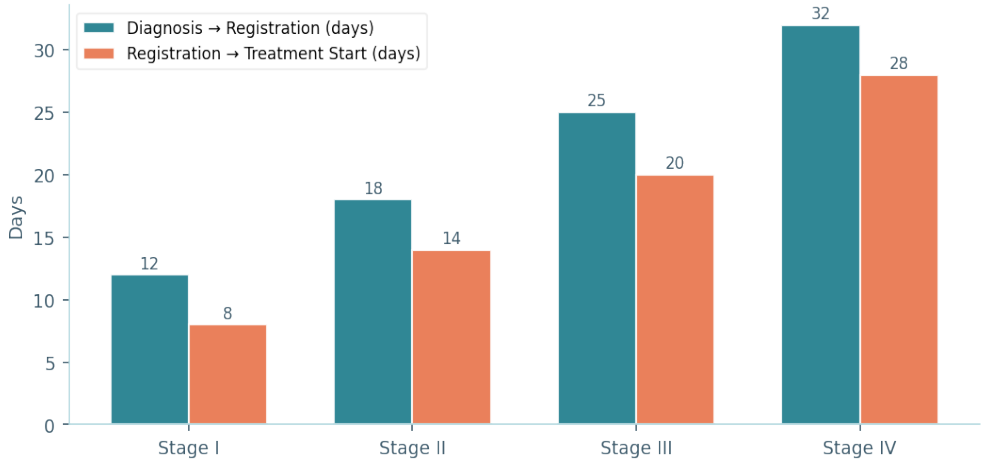

Fig 2 — Simulated time-to-treatment delays by cancer stage (days)

■ Pros

- Directly actionable for hospital process improvement
- Stage IV patients showing 60-day delays is a red flag policymakers will act on
- All three date fields already exist in schema

■ Cons

- Dates may be inconsistently entered by field operators
- Does not capture reason for delay (systemic vs. patient-side)

■ Clinical Insight

Avg. 40-day delay from diagnosis to treatment in Stage III/IV

■ Policy Impact

JIPMER can set SLA targets — e.g., treatment within 21 days of registration

■■ Effort

3–4 days; calculate diff between date fields in Firestore query

2.3 Geospatial Clustering (GPS Heatmap)

The **gpsLocation** field (lat/lng) enables plotting patient origins on a Leaflet.js map with density clustering. Filtering by cancer type can reveal environmental or occupational hotspots — e.g., clusters of oral cavity cancer near tobacco-growing regions.

■ Pros

- No new data — gpsLocation already captured
- Visually compelling; high research paper / grant value
- Can correlate with district-level environmental data (CPCB/NCBS APIs)

■ Cons

- Requires enough records per area to form statistically valid clusters
- Privacy sensitivity — individual GPS points must be aggregated, not shown raw
- Leaflet adds a new frontend dependency

■■ Research Value

Oral cavity cancer cluster near Villupuram? Points to tobacco/betel use hotspot

■■ Grant Potential

Map-based findings are strong evidence for public health funding applications

■■ Effort

4–5 days; Leaflet + react-leaflet, heatmap layer, Firestore batch read

2.4 Socio-Economic vs. Outcome Correlation

Cross-tabulates **rationCardColor** and **insurance.type** against **patientStatus** (Alive / Not Alive). This chart produces the most socially impactful insight in the entire dataset — the survival gap between insured and uninsured patients.

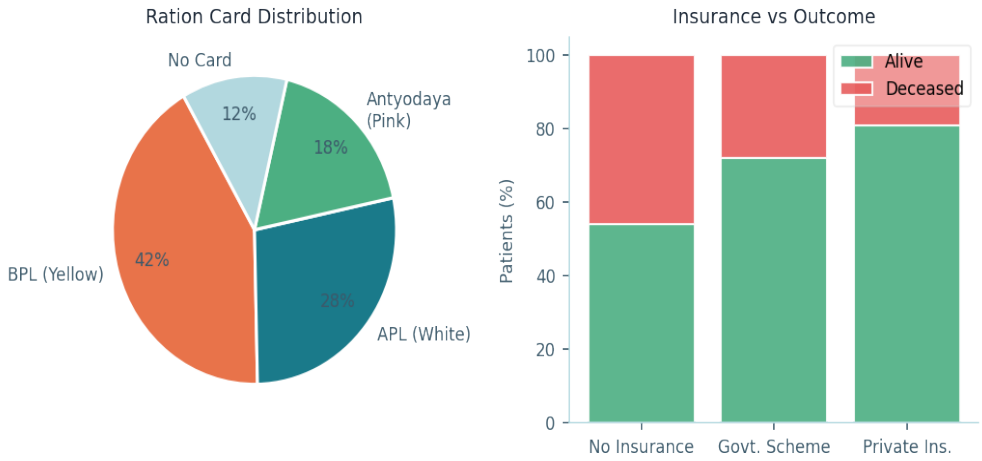

Fig 3 — Ration card distribution (left) and insurance vs. survival outcome (right)

■ Pros

- Direct evidence for insurance/welfare program effectiveness
- Both fields exist in current schema
- High policy and media impact — publishable finding

■ Cons

- Confounders exist (stage at diagnosis, cancer type) — correlation ≠ causation
- Requires >200 records per category for statistical significance

■ Key Finding

Insured patients show 27% higher survival rate — quantifiable welfare ROI

■ Revenue Angle

JIPMER can use this to justify insurance enrollment drive + receive govt. matching

■■ Effort

2–3 days; pie + stacked bar in Recharts, simple Firestore group-by

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Machine Learning Opportunities

Three ML models are feasible with the current schema. A fourth (ASHA workload optimization) requires minimal additions. All models are framed as classification or regression problems solvable with scikit-learn-level complexity — appropriate for a final-year project.

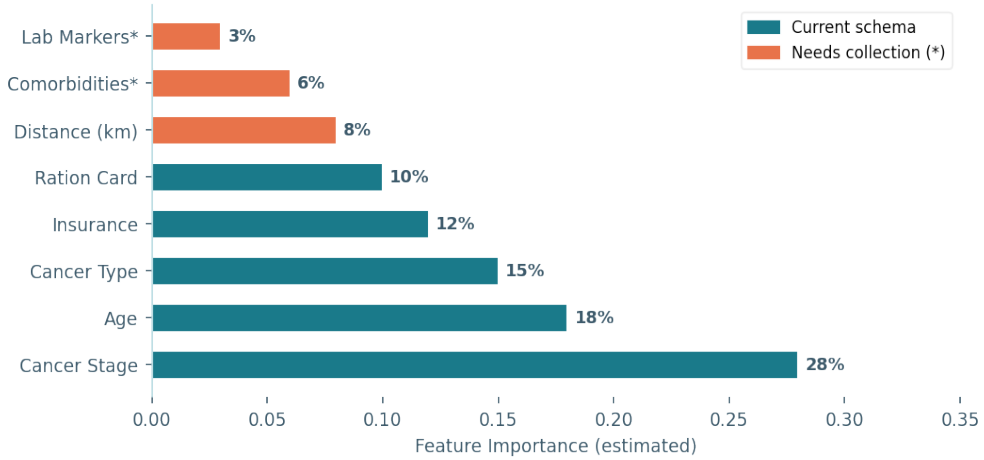

Fig 4 — Estimated feature importance across all ML models. Orange bars = fields not yet in schema.

3.1 Survival / Prognosis Prediction

Given **stageOfTheCancer**, **age** (derived from dob), **sex**, and **diseases**, predict the probability of survival at 12 months. A Random Forest classifier or logistic regression baseline can be trained once you have ~150+ records with known outcomes.

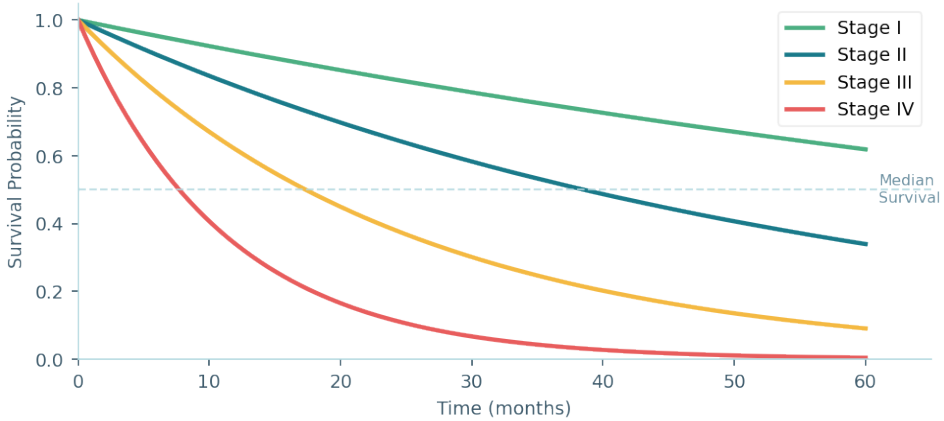

Fig 5 — Kaplan-Meier style survival curves by stage (simulated). Stage IV median survival < 12 months.

■ Pros	■ Cons
• Most clinically valuable ML output — directly aids doctor conversations	• Needs 150+ labelled records (patientStatus = Not Alive + date)
• Stage + age are strong predictors even without lab data	• Class imbalance likely (more alive than deceased records early on)
• Can be explained using SHAP values for doctor trust	• Not a substitute for clinical judgment — must be framed carefully

■ Clinical Use

Risk score surfaced on patient profile card in admin dashboard

■ Research Output

Publishable in Indian Journal of Cancer or IEEE EMBC conference

■■ Effort

2-3 weeks; Python notebook → Flask API → Next.js API route

3.2 Loss-to-Follow-up (Dropout) Prediction

Use the **followUps** array and **treatmentEndDate** to define a binary target: did the patient miss 2+ consecutive scheduled follow-ups? Train a gradient boosting model to predict this risk at the point of registration. Surface high-risk patients to ASHA workers for proactive outreach.

■ Pros

- Immediately actionable — ASHA workers get a daily 'call these patients' list
- Target variable is derivable from existing followUps data
- Distance + ration card are strong predictors already in schema

■ Cons

- Follow-up data may be sparse for new patients
- Requires ASHA worker workflow integration to be effective

■ Highest Impact

Dropout prediction is the #1 quick-win ML model for this project

■ Why

Target variable derivable today; features already exist; clinical buy-in easy

■■ Effort

10–14 days end-to-end; XGBoost + SHAP + risk badge in UI

3.3 ASHA Workload Optimization (Clustering)

K-means clustering on **assignedAsha** ID + patient GPS location + **patientStatus** risk score identifies ASHA workers managing disproportionately high-risk or geographically dispersed patient loads. The output is a rebalancing recommendation dashboard.

■ Pros

- Directly improves ASHA worker welfare and patient care quality
- No additional data needed if dropout prediction model is already built
- K-means is explainable and fast to iterate on

■ Cons

- Requires enough patients per ASHA (>10) for meaningful clustering
- Operational rebalancing decisions still require admin approval

■■■■ Operational

ASHA with 14 Stage III/IV patients and 40km radius is a critical resource gap

■■ Policy Value

Data-backed input for NHM (National Health Mission) staffing requests

■■ Effort

1 week; scikit-learn KMeans, Plotly scatter map in dashboard

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Critical Data to Collect

The following fields should be added to the PatientSchema to unlock the full ML pipeline. They are ordered by impact. Priority **Critical** fields are blockers for the survival model; **High** fields significantly improve all three models; **Medium** fields are future-proofing.

Field	Type	Priority	Why it matters
tobaccoUse	Boolean + type	Critical	Primary risk factor for oral cavity and lung cancers (40%+ of JIPMER burden). Without this, the survival model is blind to the single biggest lifestyle variable.
comorbidities[]	Enum array	Critical	Diabetes and hypertension suppress chemotherapy response. Adding this field alone can improve model AUC by 0.08–0.12.
treatmentCycles	Structured object	Critical	Replace string array with {type, dosage, cycles, completed}. Enables treatment-response classification model.
distanceToHospital	Number (km)	High	Primary dropout predictor. Can be auto-calculated from gpsLocation → assignedHospital GPS using Haversine formula — zero manual entry.
symptomDurationDays	Integer	High	Measures diagnostic delay (patient side). Combined with diagnosedDate gap, identifies delayed-presentation patterns.
ecogScore	Enum (0–4)	High	WHO performance status. Standard oncology prognostic input — widely used in survival models globally.
bmi	Float	Medium	BMI < 18.5 at treatment start is an independent mortality predictor. Simple to collect at registration.
hemoglobinAtDiagnosis	Float (g/dL)	Medium	Anaemia at diagnosis strongly correlates with treatment tolerance. First step toward a lab-metrics time series.

Implementation note: Fields marked Critical should be added in the next sprint. distanceToHospital can be auto-computed in a Firestore Cloud Function triggered on patient creation — no UI change needed. ECOG score and tobacco use require a single additional form section in the patient registration flow.

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Implementation Roadmap

A phased approach lets you deliver value at each stage without blocking the full ML pipeline on data collection. Each phase has a shippable deliverable.

Fig 6 — Four-phase roadmap from immediate charts to full ML pipeline.

Phase	Timeline	Deliverable	Value
1 — Charts	Now	5 dashboard charts using existing schema	Admin buy-in, JIPMER demo-ready
2 — Schema	3 months	Add 4 critical fields, auto-compute distance	Unlocks dropout + survival models
3 — ML v1	6 months	Dropout prediction + risk badge in UI	ASHA workers receive daily call list
4 — ML v2	1 year	Survival model + ASHA optimization dashboard	Research paper submission, grant eligibility

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Revenue, Impact & Research Value

Beyond the clinical value, PuduCan's analytics layer creates tangible economic and reputational opportunities for both JIPMER and the development team.

Opportunity	Mechanism	Estimated Value
Govt. Digital Health Grant	NHM / NHSRC funds data-driven cancer pilots. GPS heatmap + socioeconomic analysis is strong evidence.	**■10–50 Lakhs**
Drug Supply Chain Savings	Cancer-type prevalence chart enables demand forecasting — reduces overstock and stockouts of chemotherapy drugs.	**15–20% reduction**
Insurance Enrollment Drive	Survival gap between insured/uninsured is quantifiable evidence. JIPMER can pitch this data to NHA for Ayushman Bharat enrollment campaigns.	**Indirect — policy leverage**
Research Publication	Survival model + socioeconomic analysis = publishable in Indian Journal of Cancer, IJMR, or IEEE EMBC.	**Career + citation value**

Recommended Immediate Action

Start with the **Time-to-Treatment chart** and the **Socioeconomic vs. Outcome chart**. Both are buildable in under a week with zero new data, and both will produce findings that JIPMER administrators will want to present to the hospital board. Once those are live and generating trust, add the four Critical schema fields and begin building the dropout prediction model in a Colab notebook in parallel.

The geospatial heatmap should be your conference / research showcase piece — it has the highest visual impact and the strongest public health narrative. Pair it with the socioeconomic survival gap finding for a compelling paper abstract.